ADVANCED QUANTUM MECHANICS

Quantum many body physics

TECHNISCHE UNIVERSITÄT DARMSTADT

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LECTURE 13: The H_2^+ molecule

In this class and the following one we will examine two natural follow-up systems after analytically solving the electronic structure of the hydrogen atom. Before addressing the indistinguishability of electrons in the Helium atom (next lecture), we first consider the ionized molecule H_2^+, which contains only one electron and two protons. In this system, indistinguishability and quantum statistics play no role, and the goal is to show how molecular orbitals can be constructed from atomic orbitals through a linear-combination procedure.

Our strategy is as follows:

- start from the full three-body Hamiltonian with all masses and charges explicit,
- fix the center of mass at the origin and rewrite the problem in relative coordinates,
- apply the Born–Oppenheimer approximation to obtain an electronic Hamiltonian $\hat{H}_e(R)$,
- construct a molecular orbital via the Linear Combination of Atomic Orbitals (LCAO) ansatz using hydrogenic $1s$ orbitals,
- evaluate overlap and Hamiltonian matrix elements with full physical constants,
- solve the secular equation to obtain $E_g(R)$,
- include proton–proton repulsion to build the Born–Oppenheimer potential $U(R)$,
- find the equilibrium distance by solving $dU/dR = 0$.

At the end we will discuss in class (and watch videos) about some details about filling rules for electronic shells in atoms and the physics of molecular orbitals and orbital hybridization.

Solving the ionized hydrogen molecule H_2^+

We consider the molecular ion H_2^+ consisting of a single electron (mass $m = 9.11 \times 10^{-31}\,\mathrm{kg}$, charge $-e$) and two protons (mass $M = 1.67 \times 10^{-27}\,\mathrm{kg}$, charge $+e$). We will make use of the following constants and hydrogenic scales:

- Bohr radius,
$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{me^2} = 5.29 \times 10^{-11}\,\mathrm{m} \approx 0.529\,\text{Å}.$$
- Hartree energy,
$$E_{\mathrm{H}} = \frac{e^2}{4\pi\varepsilon_0 a_0} \simeq 4.36 \times 10^{-18}\,\mathrm{J} \simeq 27.2\,\mathrm{eV}.$$
- Hydrogen ground-state energy (half of Hartree energy),
$$E_{1s}^{\mathrm{H}} = -\frac{E_{\mathrm{H}}}{2} \simeq -2.18 \times 10^{-18}\,\mathrm{J} \simeq -13.6\,\mathrm{eV}.$$

Let $\mathbf{r}$ be the electron coordinate, and $\mathbf{R}_A, \mathbf{R}_B$ the proton coordinates. Owing to the symmetry of the problem, we take the center of mass in the origin and then define the relative coordinate $\mathbf{R} = \mathbf{R}_B - \mathbf{R}_A$ such that $\mathbf{R}_A = -\mathbf{R}/2$ and $\mathbf{R}_B = +\mathbf{R}/2$.

The Hamiltonian is a sum of kinetic terms (for electron and nuclei)

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}} - \frac{\hbar^2}{2M}\nabla^2_{\mathbf{R}_A} - \frac{\hbar^2}{2M}\nabla^2_{\mathbf{R}_B} + V(\mathbf{r}, \mathbf{R}_A, \mathbf{R}_B),$$

with the Coulomb potential

$$V = -\frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{|\mathbf{r}-\mathbf{R}_A|} + \frac{1}{|\mathbf{r}-\mathbf{R}_B|}\right) + \frac{e^2}{4\pi\varepsilon_0|\mathbf{R}_A - \mathbf{R}_B|},$$

including electron-proton and proton-proton contributions. Electron–proton distances then read

$$\mathbf{r}_A = \mathbf{r} - \mathbf{R}_A = \mathbf{r} + \frac{\mathbf{R}}{2}, \qquad \mathbf{r}_B = \mathbf{r} - \mathbf{R}_B = \mathbf{r} - \frac{\mathbf{R}}{2},$$

Molecular Hamiltonian in relative coordinates

As we fixed the center of mass in the origin and we will assume it separates out of the problem, we can write the full molecular Hamiltonian in terms of the relative coordinate $\mathbf{R}$ and the electronic degree of freedom $\mathbf{r}$, such that:

$$\hat{H}_{\text{rel}}(\mathbf{r}, \mathbf{R}) = \hat{T}(\mathbf{r}, \mathbf{R}) + \hat{V}_{eN}(\mathbf{r}, \mathbf{R}) + \hat{V}_{NN}(\mathbf{R}).$$

The first term is the sum of the two kinetic energy operators

$$\hat{T}(\mathbf{r}, \mathbf{R}) = -\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}} - \frac{\hbar^2}{2\mu}\nabla^2_{\mathbf{R}} \equiv \hat{T}_e + \hat{T}_N,$$

where the nuclear reduced mass is $\mu = M/2$. The potential energy terms are

$$\hat{V}_{eN} = -\frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{r_A} + \frac{1}{r_B}\right), \qquad \text{and} \qquad \hat{V}_{NN} = \frac{e^2}{4\pi\varepsilon_0\, R} = E_{\mathrm{H}}\,\frac{a_0}{R}\,.$$

The total wavefunction in this frame comes out as a solution of stationary Schrödinger equation for the two-body wavefunction

$$\hat{H}_{\text{rel}}\Psi(\mathbf{r}, \mathbf{R}) = E\,\Psi(\mathbf{r}, \mathbf{R}).$$

Born–Oppenheimer approximation -

We use the Born–Oppenheimer factorization of the molecular wavefunction

$$\Psi(\mathbf{r}, \mathbf{R}) = \phi(\mathbf{r}; \mathbf{R})\,\chi(\mathbf{R}),$$

which emphasizes that the electronic wavefunction ϕ depends only *parametrically* on $\mathbf{R}$ and is normalized for each fixed $\mathbf{R}$,

$$\int d^3r\,\left|\phi(\mathbf{r}; \mathbf{R})\right|^2 = 1.$$

Acting with the relative Hamiltonian on Ψ yields

$$\hat{H}_{\text{rel}}(\mathbf{r}, \mathbf{R})\,\Psi(\mathbf{r}, \mathbf{R}) = \chi(\mathbf{R})\,\hat{T}_e(\mathbf{r})\,\phi(\mathbf{r}; \mathbf{R}) - \frac{\hbar^2}{2\mu}\,\nabla^2_{\mathbf{R}}\big[\phi(\mathbf{r}; \mathbf{R})\,\chi(\mathbf{R})\big] + \hat{V}_{eN}(\mathbf{r}, \mathbf{R})\,\phi(\mathbf{r}; \mathbf{R})\,\chi(\mathbf{R}) + \hat{V}_{NN}(\mathbf{R})\,\phi(\mathbf{r}; \mathbf{R})\,\chi(\mathbf{R}).$$

The nuclear kinetic operator acting on the product produces

$$\nabla^2_{\mathbf{R}}\big(\phi(\mathbf{r};\mathbf{R})\,\chi(\mathbf{R})\big) = \big(\nabla^2_{\mathbf{R}}\phi\big)\,\chi + 2\,\big(\nabla_{\mathbf{R}}\phi\big)\cdot\big(\nabla_{\mathbf{R}}\chi\big) + \phi\,\nabla^2_{\mathbf{R}}\chi.$$

Introducing the electronic Hamiltonian at fixed $\mathbf{R}$,

$$\hat{H}_e(\mathbf{R}) = -\frac{\hbar^2}{2m}\,\nabla^2_{\mathbf{r}} - \frac{e^2}{4\pi\varepsilon_0}\Big(\frac{1}{r_A} + \frac{1}{r_B}\Big),$$

the full Schrödinger equation becomes

$$\chi(\mathbf{R})\,\hat{H}_e(\mathbf{R})\,\phi(\mathbf{r};\mathbf{R}) + \hat{V}_{NN}(\mathbf{R})\,\phi(\mathbf{r};\mathbf{R})\,\chi(\mathbf{R}) - \frac{\hbar^2}{2\mu}\Big[(\nabla^2_{\mathbf{R}}\phi)\,\chi + 2(\nabla_{\mathbf{R}}\phi)\cdot(\nabla_{\mathbf{R}}\chi) + \phi\,\nabla^2_{\mathbf{R}}\chi\Big] = E\,\phi(\mathbf{r};\mathbf{R})\,\chi(\mathbf{R}).$$

Projecting onto ϕ (multiply by ϕ^* and integrate over $\mathbf{r}$) gives the exact nuclear equation

$$-\frac{\hbar^2}{2\mu}\Big[\tau(R)\,\chi(R) + 2\,\boldsymbol{\alpha}(R)\cdot\nabla_{\mathbf{R}}\chi(R) + \nabla^2_{\mathbf{R}}\chi(R)\Big] + \Big[E_e(R) + V_{NN}(R)\Big]\chi(R) = E\,\chi(\mathbf{R}).$$

This is a differential equation defining the eigenvalue problem for the nuclear wavefunction. The following quantities are defined as

$$E_e(R) = \int d^3r\,\phi^*(\mathbf{r};R)\,\hat{H}_e(R)\,\phi(\mathbf{r};R), \qquad \boldsymbol{\alpha}(R) = \int d^3r\,\phi^*(\mathbf{r};R)\,\nabla_{\mathbf{R}}\phi(\mathbf{r};R),$$

$$\tau(R) = \int d^3r\,\phi^*(\mathbf{r};R)\,\nabla^2_{\mathbf{R}}\phi(\mathbf{r};R).$$

Born–Oppenheimer approximation

The Born–Oppenheimer approximation neglects the derivative couplings

$$\boldsymbol{\alpha}(R) = \int d^3r\,\phi^*(\mathbf{r};R)\,\nabla_{\mathbf{R}}\phi(\mathbf{r};R) \approx 0, \qquad \tau(R) = \int d^3r\,\phi^*(\mathbf{r};R)\,\nabla^2_{\mathbf{R}}\phi(\mathbf{r};R) \approx 0.$$

This is justified by the fact that $M \gg m$, which implies that derivatives with respect to the relative nuclear coordinate are suppressed by a factor of order m/M. The problem for the relative nuclear motion depends now only on the scalar R and can be expressed

$$-\frac{\hbar^2}{2\mu}\,\frac{1}{R^2}\,\frac{d}{dR}\Big(R^2\,\frac{d}{dR}\Big)\,\chi(R) \;+\; U(R)\,\chi(R) = E\,\chi(R), \qquad U(R) = E_e(R) + \frac{e^2}{4\pi\varepsilon_0 R}.$$

The electronic equation is then

$$\hat{H}_e(R)\,\phi(\mathbf{r};R) = E_e(R)\,\phi(\mathbf{r};R).$$

where

$$\hat{H}_e(\mathbf{R}) = -\frac{\hbar^2}{2m}\,\nabla^2_{\mathbf{r}} - \frac{e^2}{4\pi\varepsilon_0}\Big(\frac{1}{r_A} + \frac{1}{r_B}\Big),$$

Remark on non–adiabatic effects. The terms $\boldsymbol{\alpha}(R)$ and $\tau(R)$ are called *non–adiabatic couplings*. They quantify the change of the electronic state $\phi(\mathbf{r};R)$ when the nuclei move, and are responsible for coupling between different electronic potential energy surfaces. In situations where two electronic energies $E_e(R)$ approach each other (avoided crossings or conical intersections), these couplings can no longer be neglected and lead to transitions between different electronic states. In such cases the dynamics involves coupled nuclear motion on multiple potential surfaces rather than a single Born–Oppenheimer potential $U(R)$. For heavy nuclei in H_2^+ the adiabatic approximation is excellent, but in larger molecules and excited electronic configurations non–adiabatic effects play an essential role in molecular spectroscopy, chemical reaction dynamics, and photochemistry.

LCAO (Linear Combination of Atomic Orbitals) ansatz and secular equation -

We now aim at constructing the electronic wavefunction by using what we already know, the normalized atomic $1s$ orbital (as the ground state wavefunction for the electron in the hydrogen atom):

$$\phi_{1s}(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}}\, e^{-\,r/a_0}.$$

With the nuclear COM at the origin and protons at $\pm\,\mathbf{R}/2$, we define

$$\phi_A(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}}\, e^{-\,|\mathbf{r}+\mathbf{R}/2|/a_0}, \qquad \phi_B(\mathbf{r}) = \frac{1}{\sqrt{\pi a_0^3}}\, e^{-\,|\mathbf{r}-\mathbf{R}/2|/a_0}.$$

We can now write an approximate electronic wavefunction as a combination of the two electronic orbitals with coefficients that depend parametrically on the nuclear relative distance:

$$\phi(\mathbf{r}; R) = c_A(R)\,\phi_A(\mathbf{r}) + c_B(R)\,\phi_B(\mathbf{r}).$$

We now insert the LCAO ansatz into the **electronic** Schrödinger equation at fixed $R$and use the fact that $c_A(R)$ and $c_B(R)$ do not depend on $\mathbf{r}$, to pull them out of the electronic Hamiltonian:

$$c_A(R)\,\hat{H}_e(R)\phi_A(\mathbf{r}) + c_B(R)\,\hat{H}_e(R)\phi_B(\mathbf{r}) = E_e(R)\Big[c_A(R)\,\phi_A(\mathbf{r}) + c_B(R)\,\phi_B(\mathbf{r})\Big].$$

To obtain equations for $c_A(R)$ and $c_B(R)$, we project onto ϕ_A and ϕ_B by multiplying with $\phi_A^*(\mathbf{r})$ or $\phi_B^*(\mathbf{r})$ and integrating over d^3r. Projection onto ϕ_A and ϕ_B yields

$$c_A(R)\,H_{AA}(R) + c_B(R)\,H_{AB}(R) = E_e(R)\Big[c_A(R)\,\underbrace{\langle\phi_A|\phi_A\rangle}_{=1} + c_B(R)\,S(R)\Big],$$

and

$$c_A(R)\,H_{BA}(R) + c_B(R)\,H_{BB}(R) = E_e(R)\Big[c_A(R)\,S(R) + c_B(R)\,\underbrace{\langle\phi_B|\phi_B\rangle}_{=1}\Big].$$

The matrix elements are defined as

$$H_{AA}(R) = \int d^3r\,\phi_A^*(\mathbf{r})\,\hat{H}_e(R)\,\phi_A(\mathbf{r}), \qquad H_{AB}(R) = \int d^3r\,\phi_A^*(\mathbf{r})\,\hat{H}_e(R)\,\phi_B(\mathbf{r}),$$

$$H_{BA}(R) = \int d^3r\,\phi_B^*(\mathbf{r})\,\hat{H}_e(R)\,\phi_A(\mathbf{r}), \qquad H_{BB}(R) = \int d^3r\,\phi_B^*(\mathbf{r})\,\hat{H}_e(R)\,\phi_B(\mathbf{r}),$$

and the overlap integral is

$$S(R) = \int d^3r\,\phi_A^*(\mathbf{r})\,\phi_B(\mathbf{r}).$$

Eigenvalue problem using the LCAO (Linear Combination of Atomic Orbitals) ansatz

In matrix form, this is the generalized eigenvalue problem using the LCAO ansatz:

$$\begin{pmatrix} H_{AA}(R) & H_{AB}(R) \\ H_{BA}(R) & H_{BB}(R) \end{pmatrix} \begin{pmatrix} c_A(R) \\ c_B(R) \end{pmatrix} = E_e(R) \begin{pmatrix} 1 & S(R) \\ S(R) & 1 \end{pmatrix} \begin{pmatrix} c_A(R) \\ c_B(R) \end{pmatrix}.$$

The matrix elements can be analytically derived:

$$H_{AA}(R) = H_{BB}(R) = E_{\mathrm{H}} \left[-\frac{1}{2} - \frac{a_0}{R} + \exp\left(-\frac{2R}{a_0} \right) \left(1 + \frac{a_0}{R} \right) \right],$$

and

$$H_{AB}(R) = H_{BA}(R) = E_{\mathrm{H}}\, S(R) + E_{\mathrm{H}}\, \exp\left(-\frac{2R}{a_0} \right) \left(1 + \frac{a_0}{R} \right).$$

as well as

$$S(R) = \left(1 + \frac{R}{a_0} + \frac{1}{3} \frac{R^2}{a_0^2} \right) e^{-R/a_0}.$$

For symmetry reasons (if we change the indexes A with B the problem is unchanged - this is left as an exercise) the generalized eigenvalue problem has two symmetry–adapted solutions:

- **Bonding (gerade) combination**: $c_A = c_B$,

$$\phi_+(\mathbf{r}; R) \propto \phi_A(\mathbf{r}) + \phi_B(\mathbf{r}).$$

- **Antibonding (ungerade) combination**: $c_A = -c_B$,

$$\phi_-(\mathbf{r}; R) \propto \phi_A(\mathbf{r}) - \phi_B(\mathbf{r}).$$

Born–Oppenheimer potential energy surfaces -

The effective potentials for the nuclear motion (Born–Oppenheimer surfaces) are

$$U_\pm(R) = E_e^\pm(R) + \hat{V}_{NN}(R), \qquad \hat{V}_{NN}(R) = \frac{e^2}{4\pi\varepsilon_0\, R} = E_{\mathrm{H}} \frac{a_0}{R}.$$

Inserting $U_\pm(R)$ into the radial nuclear Schrödinger equation,

$$\left[-\frac{\hbar^2}{2\mu} \frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{d}{dR} \right) + U_\pm(R) \right] \chi_\pm(R) = E\, \chi_\pm(R),$$

we observe the following.

The bonding branch $U_+(R)$ possesses a minimum at some equilibrium distance R_{eq}. Therefore, the nuclear Schrödinger equation has bound-state solutions $\chi_+(R)$, corresponding to vibrational levels of a stable molecule.

The antibonding branch $U_-(R)$ is essentially repulsive and does not exhibit a potential well. Consequently, the nuclear Schrödinger equation in $U_-(R)$ has no bound nuclear eigenstates (only scattering states). Hence, although the antibonding electronic solution *exists* (it solves the electronic equation), it does not lead to a bound molecular state for the combined electron+nuclear problem.

Bonding solution -

Inserting $c_A = c_B$ into the matrix equation gives the *bonding* energy and normalized eigenvector are

$$E_e^{(+)}(R) = \frac{H_{AA}(R) + H_{AB}(R)}{1 + S(R)} \qquad \text{and} \qquad \boldsymbol{c}^{(+)}(R)) = \frac{1}{\sqrt{2(1+S(R))}} \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

such that the bonding solution is

$$\phi_+(\mathbf{r}; R) = \frac{1}{\sqrt{2(1+S(R))}} \frac{1}{\sqrt{\pi a_0^3}} \left[e^{-|\mathbf{r}+\mathbf{R}/2|/a_0} + e^{-|\mathbf{r}-\mathbf{R}/2|/a_0} \right].$$

Now we focus on the second part of the job: finding the wavefunctions for the nuclear problem. The nuclear Schrödinger equation for U_+ reads

$$\left[-\frac{\hbar^2}{2\mu} \frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{d}{dR} \right) + U_+(R) \right] \chi_+(R) = E\, \chi_+(R).$$

with

$$U_+(R) = E_e^+(R) + \frac{e^2}{4\pi\varepsilon_0 R}.$$

We will map the equation above to a quantum harmonic oscillator oscillator equation for small displacements. We remove the radial prefactor by defining

$$u(R) \equiv R\, \chi_+(R), \qquad \chi_+(R) = \frac{u(R)}{R}.$$

One verifies that

$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{du}{dR} \right) = \frac{1}{R} \frac{d^2 u}{dR^2},$$

so that multiplying the equation by R gives the ordinary one-dimensional form

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dR^2} + U_+(R)\, u(R) = E\, u(R).$$

Let R_{eq} be the minimum of $U_+(R)$,

$$\left. \frac{dU_+}{dR} \right|_{R=R_{\text{eq}}} = 0, \qquad U_+''(R_{\text{eq}}) > 0.$$

For small oscillations we expand U_+ to second order around R_{eq}:

$$U_+(R) \simeq U_+(R_{\text{eq}}) + \frac{1}{2} k \big(R - R_{\text{eq}} \big)^2, \qquad k = \left. \frac{d^2 U_+}{dR^2} \right|_{R=R_{\text{eq}}}.$$

Introducing the displacement coordinate

$$q \equiv R - R_{\text{eq}},$$

the nuclear equation becomes

$$\left[-\frac{\hbar^2}{2\mu} \frac{d^2}{dq^2} + \frac{1}{2} k\, q^2 \right] u(q) = \big(E - U_+(R_{\text{eq}}) \big)\, u(q),$$

which is the one-dimensional quantum harmonic oscillator with frequency

$$\omega_{\text{vib}} = \sqrt{\frac{k}{\mu}}.$$

For small oscillations $\chi_+(R)$ is approximately has the same harmonic shape as $u(q)/R_{\text{eq}}$. The solutions to the equation above are the usual quantum harmonic oscillator solutions.

The equilibrium distance comes out

$$R_{\text{eq}} \approx 2.00\, a_0,$$

i.e. about one nanometer scale,

$$R_{\text{eq}} \approx 1.06 \times 10^{-10}\ \text{m}.$$

Lecture summary

- **Full molecular problem**: we begin with the full electron+nuclei wavefunction $\Psi(\mathbf{r}, \mathbf{R})$ obeying

$$\hat{H}_{\text{rel}}\Psi(\mathbf{r}, \mathbf{R}) = E\,\Psi(\mathbf{r}, \mathbf{R}).$$

- **Born–Oppenheimer approximation**: factorize

$$\Psi(\mathbf{r}, \mathbf{R}) = \phi(\mathbf{r}; R)\,\chi(R),$$

  where ϕ solves the *electronic* problem at fixed R,

$$\hat{H}_e(R)\,\phi = E_e(R)\,\phi,$$

  defining the electronic energy $E_e(R)$.

- **LCAO and bonding solution**: the electronic wavefunction is approximated by a Linear Combination of Atomic Orbitals (LCAO),

$$\phi = c_A\phi_A + c_B\phi_B,$$

  leading to a 2×2 eigenvalue problem. The symmetric (gerade) solution gives the bonding state

$$\phi_+(\mathbf{r}; R) = \frac{1}{\sqrt{2(1 + S(R))}}\,\frac{1}{\sqrt{\pi a_0^3}}\left[e^{-|\mathbf{r}+\mathbf{R}/2|/a_0} + e^{-|\mathbf{r}-\mathbf{R}/2|/a_0}\right].$$

- **Born–Oppenheimer potential and nuclear motion**: adding proton–proton repulsion yields the BO potential

$$U(R) = E_e(R) + \frac{e^2}{4\pi\varepsilon_0 R},$$

  so the nuclear motion obeys

$$\left[-\frac{\hbar^2}{2\mu}\,\frac{1}{R^2}\,\frac{d}{dR}\left(R^2\frac{d}{dR}\right) + U(R)\right]\chi(R) = E\,\chi(R).$$

  Near the minimum of the bonding potential, the nuclear motion becomes approximately a harmonic oscillator, whose solutions describe vibrational levels around the equilibrium distance.

Appendix: regarding the Born-Oppenheimer approximation

Let us now make a few comments on approximations:

- **Physical meaning and hierarchy of time scales.** - The Born–Oppenheimer approximation exploits the fact that the nuclei are much heavier than the electron ($M \gg m$). Consequently, electrons adjust almost instantaneously to a given nuclear configuration. In other words, the electronic state $\psi(\mathbf{r}; R)$ depends parametrically on the slow variable R, and the electron "follows" the nuclear motion in an adiabatic manner. The nuclear kinetic energy then acts on the effective potential energy surface $U(R)$ defined by the electronic eigenvalue plus the proton–proton Coulomb repulsion.
- **Separation of energy scales.** - Typical electronic excitation energies are of the order of several eV, whereas vibrational excitations are smaller (hundreds of meV). This separation ensures that, to lowest order, the electronic

wavefunction is always close to its instantaneous ground state as the nuclei move. Only for near-degenerate electronic levels or in the presence of avoided crossings do non-adiabatic processes become relevant.

- **Non-adiabatic couplings (beyond the simplest approximation).** - A more accurate treatment keeps the nuclear kinetic operator acting on the full wavefunction $\Psi(\mathbf{r}, \mathbf{R})$ and reveals additional terms containing derivatives of $\psi(\mathbf{r}; R)$ with respect to R. These are called non-adiabatic coupling terms. They lead to corrections to the potential energy curves and can cause transitions between electronic states (molecular photodissociation, curve crossings, etc.). In molecular physics these couplings are central to vibronic phenomena.
- **Potential energy curves and molecular bonding.** - By solving the electronic equation for different R, we obtain $E_e(R)$ as a function of the internuclear separation. Its minimum defines the equilibrium bond length, and the curvature around that minimum determines the vibrational frequency. Thus, the entire problem of molecular binding is reduced to computing electronic energies as functions of fixed nuclear geometries, a key concept in quantum chemistry.
- **Born–Oppenheimer vs. full molecular Hamiltonian.** - In the full Hamiltonian electrons and nuclei are strongly coupled through the Coulomb interaction, and no exact separation of variables exists. The Born–Oppenheimer approach systematically replaces the full six-dimensional (plus nuclear) eigenproblem by a sequence of much simpler electronic problems at fixed geometry, followed by a one-dimensional nuclear motion on the resulting potential. The approximation is not exact, but it is extremely accurate in most stable molecules under ordinary conditions.